

Paper Review

Half-trek criterion for generic identifiability of linear structural equation models.

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· Foygel, R., Draisma, J., & Drton, M. (2012).

Half-trek criterion for generic identifiability of linear structural equation models.

The Annals of Statistics, 1682-1713.

Main contributions and outline

Main contributions

- Propose a **new graphical condition** that is sufficient for **generic identifiability** in a **mixed graph generated by linear SEM**.
- Develop a **polynomial-time algorithm** to test whether the condition is satisfied.
- Provide various simulations along with exhaustive algebraic computations to verify the criteria.

- Introduction
- Preliminaries
- Main identifiability and nonidentifiability results
- Efficient algorithms for HTC-classification
- Computational experiments
- Discussion

Introduction

Introduction

- This study copes with an identifiability of **linear SEM with allowing errors to be correlated**.
- The statistical literature often refers to this as **almost-everywhere identifiability**.
- **G-criterion** proposed by Brito and Pearl (2006) is a well-known trial to tackle the issue.
- This paper provides a new identifiability condition, named as **the half-trek criterion**, which **captures a wider range of linear SEMs than the G-criterion**.
- The half-trek criterion is efficiently checkable in **polynomial-time algorithm**.

Motivating example



$\mathcal{G}$: Three-node MG

- Consider a three-node linear SEMs:

$$X_1 = \epsilon_1, \quad X_2 = \lambda_{12}X_1 + \epsilon_2, \quad X_3 = \lambda_{23}X_2 + \epsilon_3,$$

$$\text{with } \text{Var}(\epsilon) = \begin{pmatrix} \omega_{11} & 0 & 0 \\ 0 & \omega_{22} & \omega_{23} \\ 0 & \omega_{23} & \omega_{33} \end{pmatrix}.$$

- Its inverse covariance matrix is as follows:

$$\begin{aligned} \Sigma^{-1} &= (\sigma_{vw}) \\ &= \begin{pmatrix} \omega_{11} & \omega_{11}\lambda_{12} & \omega_{11}\lambda_{12}\lambda_{23} \\ - & \omega_{22} + \omega_{11}\lambda_{12}^2 & \omega_{23} + \lambda_{23}\sigma_{22} \\ - & - & \omega_{33} + 2\omega_{23}\lambda_{23} + \lambda_{23}^2\sigma_{22} \end{pmatrix}. \end{aligned}$$

- We can recover Λ from Σ using that

$$\lambda_{12} = \frac{\sigma_{12}}{\sigma_{11}}, \quad \lambda_{23} = \frac{\sigma_{13}}{\sigma_{12}}.$$

Preliminaries

Mixed graph

- A *mixed graph* is a triple $\mathcal{G} = (V, D, B)$, where V is a finite set of nodes and $D, B \subset V \times V$ are two sets of edges.
 - The pairs (v, w) in the set D represent **directed edges**.
 - The pairs in B are **bidirected edges** $v \leftrightarrow w$, that is $(v, w) \in B$ if and only if $(w, v) \in B$.
- If the **directed part** (V, D) **does not contain directed cycles**, then the mixed graph G is said to be *acyclic*.
- A *path* from node v to node w in a mixed graph $G = (V, D, B)$ is a sequence of edges, **each from either D or B** , that connect the consecutive nodes in a sequence of nodes beginning at v and ending in w .
- A path π from v to w is a *directed path* if **all its edges are directed and pointing to w** , that is,

$$\pi : V = V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r = W.$$

- A *trek* from source v to target w is a path from v to w whose **consecutive edges do not have any colliding arrowheads**, that is,

$$\pi : v = v_\ell^L \leftarrow v_{\ell-1}^L \leftarrow \dots \leftarrow v_1^L \leftarrow v_0^L \longleftrightarrow v_0^R \rightarrow v_1^R \dots \rightarrow v_{r-1}^R \rightarrow v_r^R = w,$$

or

$$\pi : v = v_\ell^L \leftarrow v_{\ell-1}^L \leftarrow \dots \leftarrow v_1^L \leftarrow v^T \rightarrow v_1^R \dots \rightarrow v_{r-1}^R \rightarrow v_r^R = w.$$

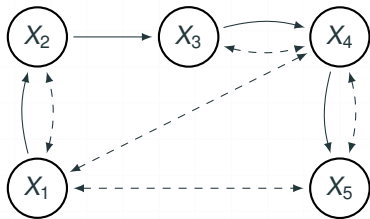
- In the first case, we denote $\text{Left}(\pi) = \{v_0^L, \dots, v_\ell^L\}$ and $\text{Right}(\pi) = \{v_0^R, \dots, v_r^R\}$.
- In the second case, we denote $\text{Left}(\pi) = \{v^T, \dots, v_\ell^L\}$ and $\text{Right}(\pi) = \{v^T, \dots, v_r^R\}$.
- Note that v itself can be the top node **if there is a directed path from v to w** , i.e., $\text{Left}(\pi) = \{v\}$.
- A trek from v to v is called the *trivial trek*.

- A *half-trek* π is a trek with $|\text{Left}(\pi)| = 1$, i.e.,

$$\pi : v \leftrightarrow v_0^R \rightarrow v_1^R \rightarrow \dots \rightarrow v_{r-1}^R \rightarrow v_r^R = w,$$

or

$$\pi : v \rightarrow v_1^R \rightarrow \dots \rightarrow v_{r-1}^R \rightarrow v_r^R = w.$$



$\mathcal{G}$: Five-node MG

- $\pi : 2 \rightarrow 3 \rightarrow 4 \leftarrow 3$ is not a trek, due to the collider node 4.
- $\pi : 2 \leftarrow 1 \leftrightarrow 4 \rightarrow 5$ is a trek, but not a half trek.
- $\pi : 1 \rightarrow 2 \rightarrow 3$ is a half-trek.

Trek system

- Consider n treks $\pi_i : i = 1, 2, \dots, n$, where x_i and y_i are the source and the target of π_i , respectively.
- If $x_i \neq x_j$ and $y_i \neq y_j$ for all $i \neq j$, we say that $\Pi = \{\pi_1, \dots, \pi_n\}$ is a *trek system* from $X = \{x_1, \dots, x_n\}$ to $Y = \{y_1, \dots, y_n\}$, denoted as $\Pi : X \rightrightarrows Y$.
- Π is a *half-trek system* if every trek π_i is a half-trek.
- The system Π is said to have *no sided intersection* if

$$\text{Left}(\pi_i) \cap \text{Left}(\pi_j) = \emptyset = \text{Right}(\pi_i) \cup \text{Right}(\pi_j) \quad \forall i \neq j.$$

- The set $\Pi = \{\pi_1, \pi_2\}$ comprising the two treks

$$\pi_1 : 1 \leftrightarrow 4, \quad \pi_2 : 3 \rightarrow 4 \rightarrow 5$$

has a sided intersection since $\text{Right}(\pi_1) \cup \text{Right}(\pi_2) = \{4\}$.

Definition 1. (Linear structural equation model)

The linear structural equation model given by a mixed graph $\mathcal{G} = (V, D, B)$ on $V = [m]$ is the family of all m -variate normal distributions with covariance matrix

$$\Sigma = (I - \Lambda)^{-T} \Omega (I - \Lambda)^{-1}$$

for $\Lambda \in \mathbb{R}_{reg}^D$ and $\Omega \in PD(B)$.

- $\mathbb{R}_{reg}^D$ is the set of real $m \times m$ -matrices Λ with support D such that $I - \Lambda$ is invertible.
- $PD(B)$ is the set of positive definite symmetric $m \times m$ -matrices Ω with support B , that is, $\omega_{vw} = 0$ if $v \neq w$ and $v \leftrightarrow w \notin B$.

Identifiability

- For ease of notation, let $\Theta = \mathbb{R}_{reg}^D \times PD(B)$.
- The *parameterization map* $\phi_{\mathcal{G}}$ is defined as

$$\phi_{\mathcal{G}} : (\Lambda, \Omega) \mapsto (I - \Lambda)^{-T} \Omega (I - \Lambda)^{-1}.$$

- **Identifiability holds if $\phi_{\mathcal{G}}$ is injective on Θ , or a suitably large subset.**
- Define the *fiber* $\mathcal{F}$ as

$$\mathcal{F}(\Lambda, \Omega) = \{(\Lambda', \Omega') \in \Theta \setminus S : \phi_{\mathcal{G}}(\Lambda', \Omega') = \phi_{\mathcal{G}}(\Lambda, \Omega)\},$$

where $S \subset \Theta$ is a null set.

- Put differently, **verifying identifiability is equivalent to asking whether**
 $\mathcal{F}(\Lambda, \Omega) = \{(\Lambda, \Omega)\}.$

Generic identifiability

Definition 2. (Generic identifiability)

The mixed graph $\mathcal{G}$ is said to be *generically identifiable* if $\phi_{\mathcal{G}}$ is injective on the complement $\Theta \setminus S$ of a proper (i.e., strict) algebraic subset $S \subset \Theta$.

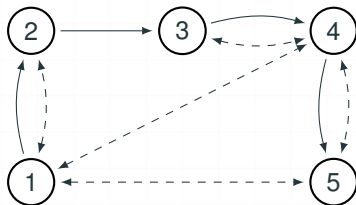
Definition 3. (Rational identifiability)

The mixed graph $\mathcal{G}$ is said to be *rationally identifiable* if there exists a proper algebraic subset $S \subset \Theta$ and a rational map ψ such that $\psi \circ \phi_{\mathcal{G}}(\Lambda, \Omega) = (\Lambda, \Omega)$ for all $(\Lambda, \Omega) \in \Theta \setminus S$.

- A proper algebraic subset has smaller dimension than Θ , and thus it has measure zero.
- In essential, **Definition 2 is about almost-everywhere identifiability.**
- Definition 3 formulates the situation where (Λ, Ω) can be recovered from Σ by some rational map ψ .
- **Rational identifiability implies generic identifiability.**
- If the fiber has positive dimension, we say $\mathcal{G}$ is *generically infinite-to-one*.

Main identifiability and nonidentifiability results

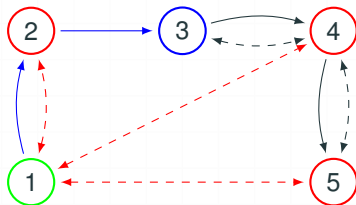
Notations



$\mathcal{G}$: Five-node MG

- $\text{pa}(v) = \{w : w \rightarrow v \in D\}$, and $\text{sib}(v) = \{w : w \leftrightarrow v \in B\}$.
- $\text{htr}(v)$ is the set of nodes in $V \setminus (\{v\} \cup \text{sib}(v))$ that can be reached from v via a half-trek.
- Put differently, a node $w \neq v$ that is not a sibling of v is in $\text{htr}(v)$ if w is a **proper descendant of v or one of its siblings**.

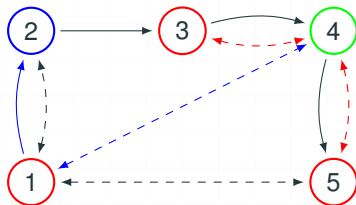
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- Put differently, a node $w \neq v$ that is not a sibling of v is in $\text{htr}(v)$ if w is a **proper descendant of v or one of its siblings**.

Half-trek criterion identifiability

Definition 4. (Half-trek criterion)

A set of nodes $Y_v \subset V$ satisfies the half-trek criterion with respect to node $v \in V$ if

- (i) $|Y_v| = |\text{pa}(v)|$,
- (ii) $Y_v \cap (\{v\} \cup \text{sib}(v)) = \emptyset$, and
- (iii) there is a system of half-treks with no sided intersection from Y_v to $\text{pa}(v)$.

Theorem 1. (HTC-identifiability)

Let $(Y_v : v \in V)$ be a family of subsets of the vertex set V of a mixed graph $\mathcal{G}$. If, for each node v , **the set Y_v satisfies the half-trek criterion with respect to v , and there is a total ordering $w < v$ whenever $w \in Y_v \cap \text{htr}(v)$** , then $\mathcal{G}$ is **rationally identifiable**.

Proof sketch of Theorem 1

Using mathematical induction, we assume that for all $w < v$, **we have recovered the entries of the vector** $\Lambda_{\text{pa}(w),w}$ as rational expressions in Σ .

To solve for $\Lambda_{\text{pa}(v),v}$, let $Y_v = \{y_1, \dots, y_n\}$ and $\text{pa}(v) = \{p_1, \dots, p_n\}$. Define $A \in \mathbb{R}^{n \times n}$ as

$$A_{ij} = \begin{cases} [(I - \Lambda)^T \Sigma]_{y_i p_j} & \text{if } y_i \in \text{htr}(v), \\ \Sigma_{y_i p_j} & \text{o.w.} \end{cases}$$

Define $b \in \mathbb{R}^n$ as

$$b_i = \begin{cases} [(I - \Lambda)^T \Sigma]_{y_i v} & \text{if } y_i \in \text{htr}(v), \\ \Sigma_{y_i v} & \text{o.w.} \end{cases}$$

By the assumption, we know the exact form of A and b . Additionally, we can show that A is generically invertible and thus

$$\Lambda_{\text{pa}(v),v} = A^{-1}b$$

holds. Subsequently, Ω is attained by calculating $\Omega = (I - \Lambda)^T \Sigma (I - \Lambda)$. $\square$

Half-trek criterion nonidentifiability

Theorem 2. (HTC-nonidentifiability)

Suppose $\mathcal{G}$ is a mixed graph in which every family $(Y_v : v \in V)$ of subsets of the vertex set V either contains a set Y_v that **fails to satisfy the half-trek criterion** with respect to v or **contains a pair of sets (Y_v, Y_w) with $v \in Y_w$ and $w \in Y_v$** . Then the parameterization $\phi_{\mathcal{G}}$ is **generically infinite-to-one**.

- Under the given conditions, the dimension of $\text{Image}(\phi_{\mathcal{G}})$ is smaller than that of $\binom{m+1}{2}$ -dimensional space of symmetric $m \times m$ matrices.
- As a result, for generic choices of $(\Lambda, \Omega) \in \Theta$, **the fiber $\mathcal{F}(\Lambda, \Omega)$ has positive dimension**.

Some propositions

Proposition 1. (HTC-identifiability of a simple acyclic mixed graph)

If $\mathcal{G}$ is a simple acyclic mixed graph, then $\mathcal{G}$ is HTC-identifiable.

Proposition 2. (HTC-nonidentifiability of a dense graph)

If a mixed graph $\mathcal{G} = (V, D, B)$ with $V = [m]$ has $|D| + |B| \geq \binom{m}{2}$ edges, then $\mathcal{G}$ is HTC-infinite-to-one.

- A mixed graph $\mathcal{G} = (V, D, B)$ is simple if there is at most one edge between any pair of nodes.
- It is straightforward that the map $\phi_{\mathcal{G}}$ cannot have a full-rank if its domain is large.

Efficient algorithms for HTC-classification

Background on the max-flow problem

- Let $\mathcal{G} = (V, D)$ be a directed graph with designated source and sink nodes $s, t \in V$.
- $c_V : V \rightarrow \mathbb{R}_{\geq 0}$ is a *node-capacity function*, and $c_D : D \rightarrow \mathbb{R}_{\geq 0}$ is an *edge-capacity function*.
- A *flow* f on $\mathcal{G}$ is a function $f : D \rightarrow \mathbb{R}_{\geq 0}$ that satisfies

$$\sum_u f(u, v) = \sum_w f(v, w) \leq c_V(v)$$

for all nodes $v \neq s, t$, and

$$f(u, v) \leq c_D(u, v)$$

for all edges $u \rightarrow v \in D$.

- The *size* $|f|$ of a flow f on $\mathcal{G}$ is the **total amount of flow passing from the source s , or to the sink t** , that is,

$$|f| := \sum_w f(s, w) = \sum_u f(u, t).$$

- The *max-flow problem* on $(\mathcal{G}, s, t, c_V, c_D)$ is the problem of **finding a flow f whose size $|f|$ is maximum**.

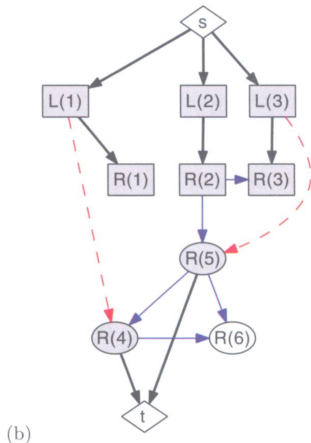
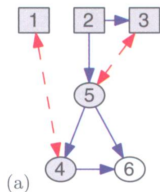
Max-flow problem and HTC-identifiability

Problem 1. (How to determine HTC-identifiability)

Given a node $v \in V$, and a subset of “allowed” nodes $A \subset V \setminus (\{v\} \cup \text{sib}(v))$, how can we efficiently determine whether there exists a subset $Y_v \subset A$ satisfying the half-trek criterion with respect to v ?

- The meaning of “allowed” will be described later.
- Answering this question is equivalent to **solving a max-flow problem** on a network $\mathcal{G}_{\text{flow}}(v, A)$ with at most $2|V| + 2$ nodes and at most $3|V| + |D| + |B|$ edges.
- In general, the computational complexity of the max-flow problem in $\mathcal{G}(V, D)$ is known to be $O((|V| + r)^3)$, where $r \leq |D|/2$ is the number of reciprocal edge pairs.

Construction of $\mathcal{G}_{\text{flow}}$



$$\mathcal{G}_{\text{flow}}(v = 6, A = \{1, 2, 3\})$$

- The vertex set of $\mathcal{G}_{\text{flow}}(v, A)$ comprises three types of nodes:
 - (a) a source s and a sink t ,
 - (b) a “left-hand copy” $L(a)$ for each $a \in A$, and
 - (c) a “right-hand copy” $R(w)$ for each $w \in V$.
- The edges of $\mathcal{G}_{\text{flow}}(v, A)$ are given by the following:
 - (i) $s \rightarrow L(a)$ and $L(a) \rightarrow R(a)$ for each $a \in A$,
 - (ii) $L(a) \rightarrow R(w)$ for each $a \leftrightarrow w \in B$,
 - (iii) $R(w) \rightarrow R(u)$ for each $w \rightarrow u \in D$,
 - (iv) $R(w) \rightarrow t$ for each $w \in \text{pa}(v)$.

Construction of $\mathcal{G}_{\text{flow}}$

- Vertex rule:
 - (a) a source s and a sink t ,
 - (b) a “left-hand copy” $L(a)$ for each $a \in A$, and
 - (c) a “right-hand copy” $R(w)$ for each $w \in V$.
- Edge rule:
 - (i) $s \rightarrow L(a)$ and $L(a) \rightarrow R(a)$ for each $a \in A$,
 - (ii) $L(a) \rightarrow R(w)$ for each $a \leftrightarrow w \in B$,
 - (iii) $R(w) \rightarrow R(u)$ for each $w \rightarrow u \in D$,
 - (iv) $R(w) \rightarrow t$ for each $w \in \text{pa}(v)$.
- A half-trek of the form $y \rightarrow x_1 \rightarrow \dots \rightarrow p$, and $y \leftrightarrow x_1 \rightarrow \dots \rightarrow p$ with $y \in A$ and $p \in \text{pa}(v)$, will appear in the flow network as

$$s \xrightarrow{(i)} L(y) \xrightarrow{(i)} R(y) \xrightarrow{(iii)} R(x_1) \xrightarrow{(iii)} \dots \xrightarrow{(iii)} R(p) \xrightarrow{(iv)} t,$$

and

$$s \xrightarrow{(i)} L(y) \xrightarrow{(ii)} R(x_1) \xrightarrow{(iii)} \dots \xrightarrow{(iii)} R(p) \xrightarrow{(iv)} t,$$

respectively.

- Additionally, all edges have capacity ∞ .
- The source s and sink t have capacity ∞ , and all other nodes have capacity 1.

Determining HTC-identifiability

Theorem 6. (Max-flow problem and half-trek criterion)

Given a mixed graph $\mathcal{G} = (V, D, B)$, a node $v \in V$ and a subset of “allowed” nodes $A \subset V \setminus (\{v\} \cup \text{sib}(v))$, there exists a set $Y_v \subset A$ satisfying the half-trek criterion with respect to v if and only if the flow network $\mathcal{G}_{\text{flow}}(v, A)$ **has maximum flow equal to $|\text{pa}(v)|$** .

Theorem 1. (HTC-identifiability)

Let $(Y_v : v \in V)$ be a family of subsets of the vertex set V of a mixed graph $\mathcal{G}$. If, for each node v , **the set Y_v satisfies the half-trek criterion with respect to v** , and **there is a total ordering $w < v$ whenever $w \in Y_v \cap \text{htr}(v)$** , then $\mathcal{G}$ is **rationally identifiable**.

- Of primary concern to achieve HTC-identifiability is that the set A should be carefully chosen to **avoid the situation where $w \in Y_v \cap \text{htr}(v)$ but $w \neq v$** .
- Define A as follows:

$$A = [\{w : w < v\} \cup (V \setminus \text{htr}(v))] \setminus [\{v\} \cup \text{sib}(v)] .$$

Algorithm with $O(|V|^2 (|V| + r)^3)$

Algorithm 1 Testing HTC-identifiability of a mixed graph

Input: $G = (V, D, B)$, a mixed graph on m nodes

Initialize: $\text{SolvedNodes} \leftarrow \{v : \text{pa}(v) = \emptyset\}$.

repeat

for $v = 1, 2, \dots, m$ **do**

if $v \notin \text{SolvedNodes}$ **then**

$A \leftarrow (\text{SolvedNodes} \cup (V \setminus \text{htr}(v))) \setminus (\{v\} \cup \text{sib}(v))$.

if $\text{MaxFlow}(G_{\text{flow}}(v, A)) = |\text{pa}(v)|$ **then**

$\text{SolvedNodes} \leftarrow \text{SolvedNodes} \cup \{v\}$.

end if

end if

end for

until $\text{SolvedNodes} = V$ or no change has occurred in the last iteration.

Output: “yes” if $\text{SolvedNodes} = V$, “no” otherwise.

Computational experiments

Exhaustive computations on small graphs

- All possible isomorphism classes of graphs with respect to permutation of the vertex set $V = [m]$ for $m \leq 5$ are simulated.
- We can single out the graphs with more than $\binom{m}{2}$ edges by Proposition 2.
- The remaining groups are classified into three disjoint groups:
 - Rationally identifiable graphs (HTC-identifiable),
 - generically infinite-to-one graphs (HTC-infinite-to-one), and
 - generically finite-to-one graphs.
- Each graph is categorized using algebraic techniques in advance.

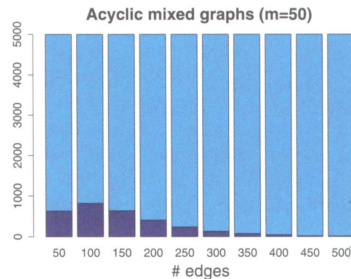
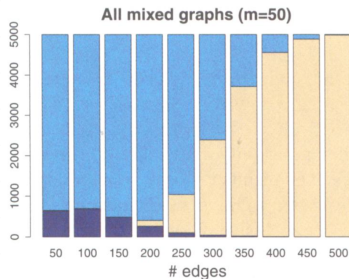
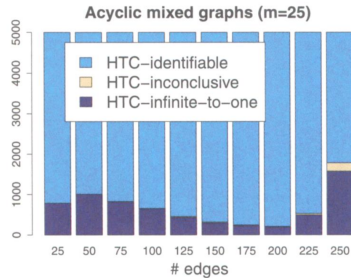
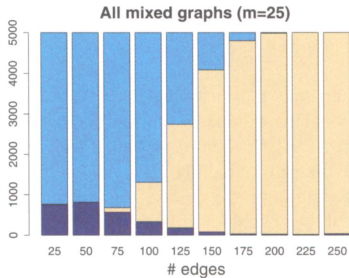
Exhaustive computations on small graphs

Unlabeled mixed graphs	$m = 3$		$m = 4$		$m = 5$	
	Total	HTC	Total	HTC	Total	HTC
Acyclic, $\leq \binom{m}{2}$ edges	22		715		103,670	
rationally identifiable	17	17	343	343	32,378	32,257
generically finite-to-one	0	—	4	—	1166	—
generically ∞ -to-one	5	5	368	368	70,126	70,099
Acyclic, $> \binom{m}{2}$ edges	18		852		152,520	
Cyclic, $\leq \binom{m}{2}$ edges	6		718		348,175	
rationally identifiable	2	2	239	230	91,040	78,586
generically finite-to-one	1	—	75	—	44,703	—
generically ∞ -to-one	3	3	404	383	212,432	202,697
Cyclic, $> \binom{m}{2}$ edges	58		9307		8,439,859	

Simulations for large graphs

- Algebraic computations are not feasible for larger graphs.
- Alternatively, HTC-status of randomly generated mixed graphs with $m = 25$ and $m = 50$ nodes are tested.
- 5,000 labeled mixed graphs on m nodes with n edges are randomly sampled.
 - For each value $n = k \cdot m$ for $k \in [10]$, the edge set of size n is chosen from the set of all possible edges while maintaining acyclic.

Simulations for large graphs



Discussion

- **Identifiability of correlated linear SEM** is discussed.
- The criteria can be checked in **polynomial-time**.
- The half-trek criterion is **weaker than G-criterion**.
- There still remains a “**gap**” **between HTC-identifiability and HTC-infinite-to-one**.

Reference

- Brito, C. and Pearl, J. (2002a). A new identification condition for recursive models with correlated errors. *Struct. Equ. Model.* **9** 459-474. MR1930449